
Si3050/52/54/56 LAYOUT GUIDELINES

1. Description

The Si305x chipsets improve upon Skyworks Solutions' groundbreaking family of silicon direct access arrangement (DAA) products by providing the highest integration, lowest cost, and smallest area DAA solution for modem and voice applications. Skyworks Solutions' DAA chipsets consist of two devices. These two integrated circuits are commonly referred to as the digital-side device and the line-side device. This nomenclature refers to which portion of the circuit each chip interfaces: the digital logic of the system or the telephone line. When designing a board with the Si305x, there are several layout considerations that will assist the printed circuit board designer in developing a product that meets the target application's requirements. This document provides guidelines for board design layout, review, and modification using the Si305x chipsets.

1.1. Form Factor Overview

The layout guidelines for the Si3050/52/54/56 are similar, but there are some differences depending on form factor or application. Typically, PCI, CNR, and AMR designs are larger board area designs and have more freedom for different component packages (i.e., thru-hole versus surface mount) and spacing than other types of designs. PCI, CNR, and AMR guidelines have been grouped together except where there are form factor specific requirements. MiniPCI and MDC form factors are much smaller and typically have limited options for component packages and restrictive spacing requirements. Finally, the Si3050 is typically used in applications where channel density is at a premium. Therefore, the Si3050 layout has been optimized to allow for multiple channels to be placed adjacent to each other in a customer design. The Si3050 digital-side device is available in the TSSOP-20 or QFN-24 package, and the associated line-side device is available in the TSSOP-16, SOIC-16, or QFN-20

package. The Si3050 schematic diagrams and layout examples included in this document refer to the TSSOP packages for both the system-side and the line-side devices. Although the pin numbers for specific signals do change between packages, the relative ordering of the signals is the same for the different package options, and the layout guidelines presented herein are generally applicable regardless of which package option is selected.

1.2. Placement Guidelines

Figure 27 on page 25 shows the typical application circuit for the line-side circuit of the DAA. Note that, for Si3050 designs, it is recommended that zero-ohm resistors R15 and R16 be replaced with ferrite beads in order to decrease emissions (see Si3050+Si3011/18/19 data sheet). Figures 1–4 depict typical placement of the chipset, discrete components, and the RJ11 connector for the Si3050, Si3052 PCI, Si3054 CNR, and Si3054 MDC. Note the placement of U1 (the Si305x digital-side chip) and U2 (the Si3018/Si3019 line-side chip). U1, U2, C1, and C2 should be placed so that the distance from U1 to U2 through the C1, C2 capacitors is less than 2 inches. Place R12 and R13 near the C1A, C2A pins on the Si305x. Use a minimum of 15 mil thick traces for this connection. Aligning the Si3050/54/56 digital-side chip so that the C1A/C2A pins face pins 1–8 of the line-side chip makes some of the other specific layout guidelines easier to implement.

With the Si3052 digital-side chip, the C1A/C2A pins are typically not aligned with the C1B/C2B pins on the line-side device. This alignment is a result of having to route the large number of PCI lines to the PCI edge connector. Figure 6 on page 4 shows a typical routing of the C1A/C2A lines for the Si3052.

Utilizing this placement also allows the design to closely resemble the Skyworks Solutions' example layouts, enabling the designer to directly follow these guidelines.

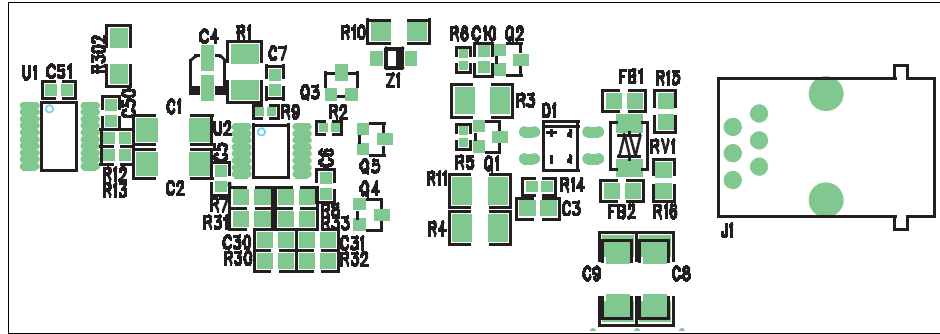


Figure 1. Si3050 Placement

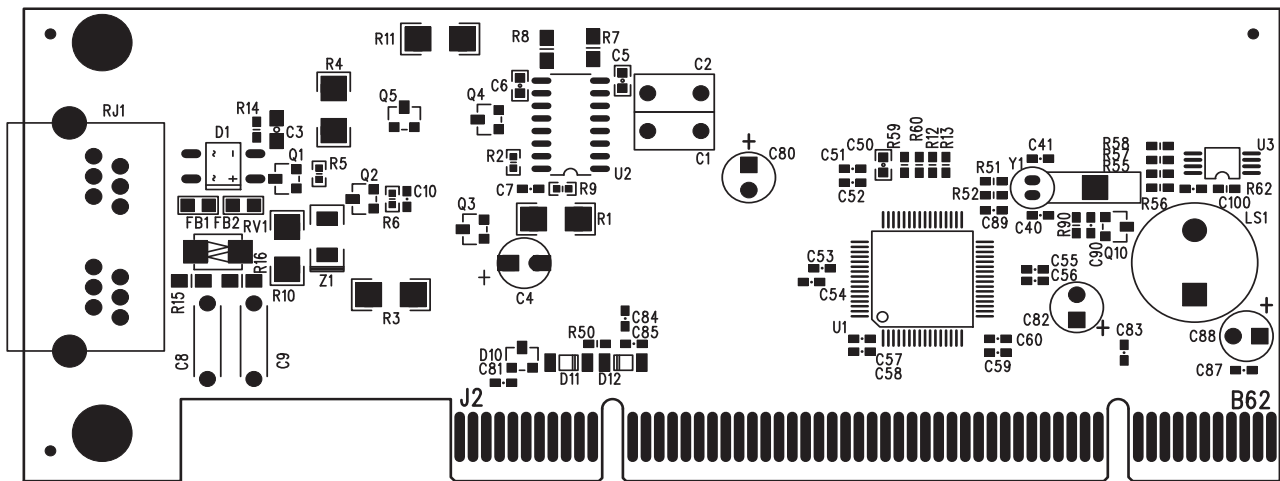


Figure 2. Si3052 PCI Placement

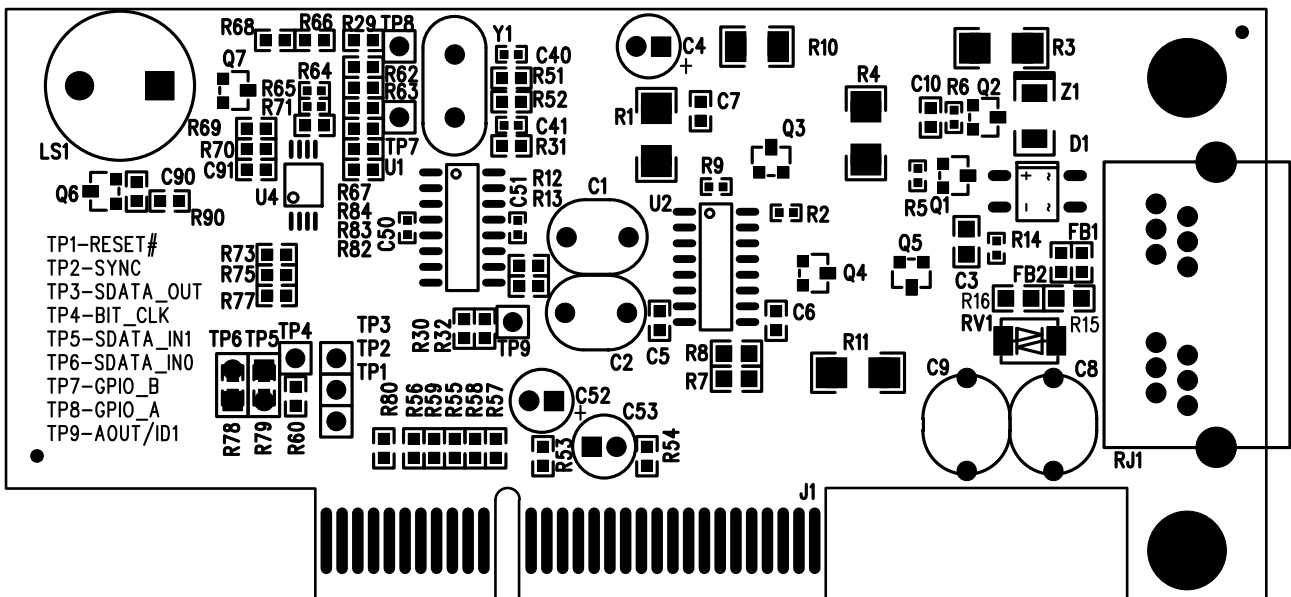


Figure 3. Si3054 CNR Placement

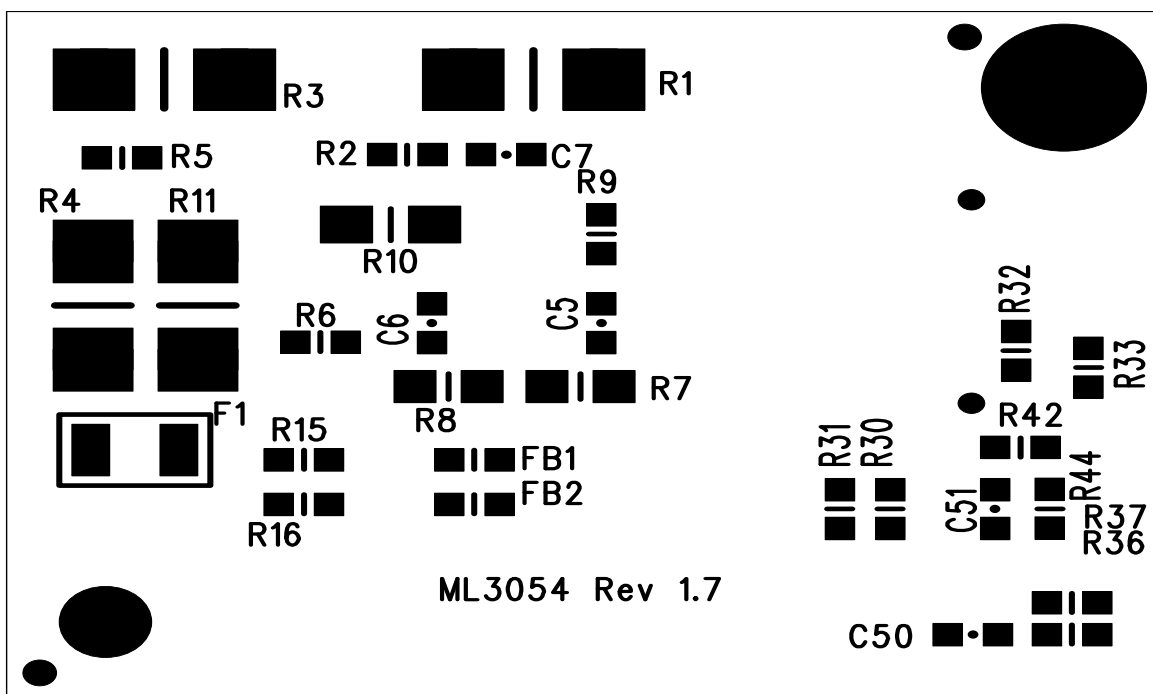


Figure 4. Si3054 MDC Side 1 Placement

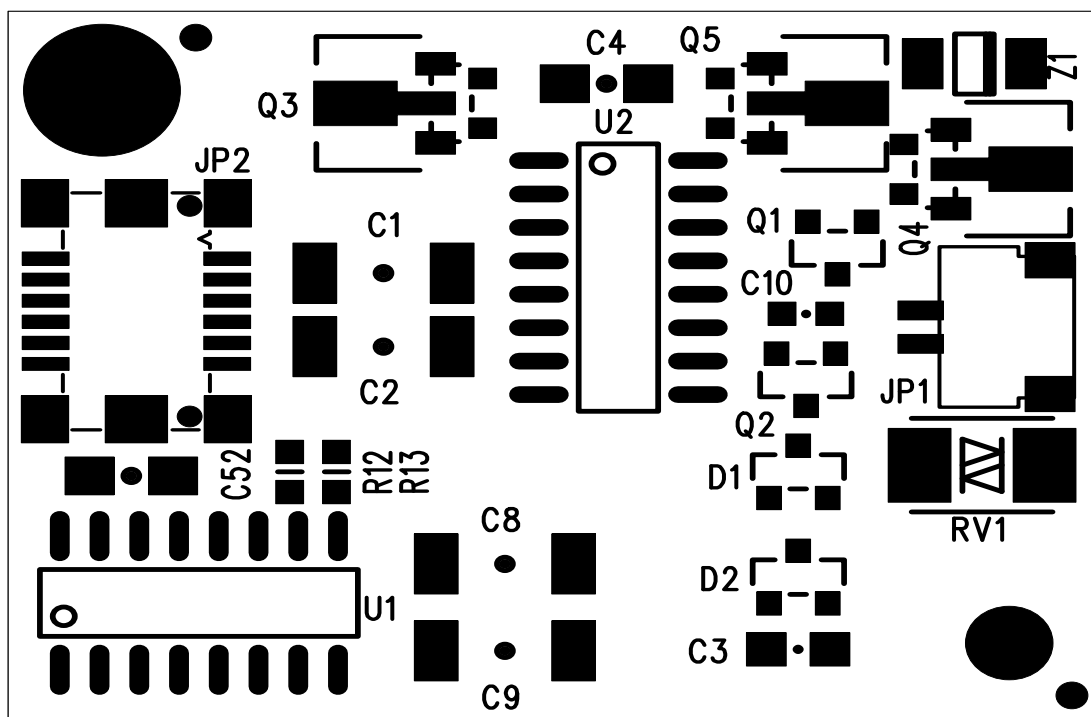


Figure 5. Si3054 MDC Side 2 Placement

1.3. Isolation Barrier Creepage Spacing Considerations

A device that connects to the telephone network interfaces to an environment where high-voltage ring events, transients, and even lightning-induced surges can occur. One of the principal purposes of a DAA is to provide isolation of a higher voltage interface, in this case the telephone line, from a lower voltage interface, or ground-referenced integrated circuitry. In the telecom industry, these areas of the circuit are commonly referred to as telephone network voltage (TNV) for the high-voltage interface, and safety extra-low voltage (SELV) for the low-voltage interface. The spacing between high-voltage components and traces and low-voltage components and traces is commonly referred to as the isolation barrier.

The highlighted area of the modem in Figure 6 is the isolation barrier.

The Skyworks Solutions DAA utilizes high-voltage capacitors to maintain isolation while providing a means of communication across the isolation barrier. In addition, spacing is required between the high-voltage components to the low-voltage components in order to meet several telecom, EMC, and safety specifications. For more details on the various safety requirements, consult the Skyworks Solutions application note "AN17: Designing for International Safety Compliance" as well as your own safety experts and consultants.

Skyworks Solutions recommends the following spacing guidelines and capacitor selection for the TNV to SELV isolation barrier, grouped by application.

PCI/CNR/AMR/ACR Modem: Use through-hole Y2 class capacitors for the C1, C2, C8, and C9 components, and ensure a 5 mm wide isolation barrier from TNV to SELV. A properly implemented design using these guidelines should provide a minimum of 5 kV of longitudinal surge immunity.

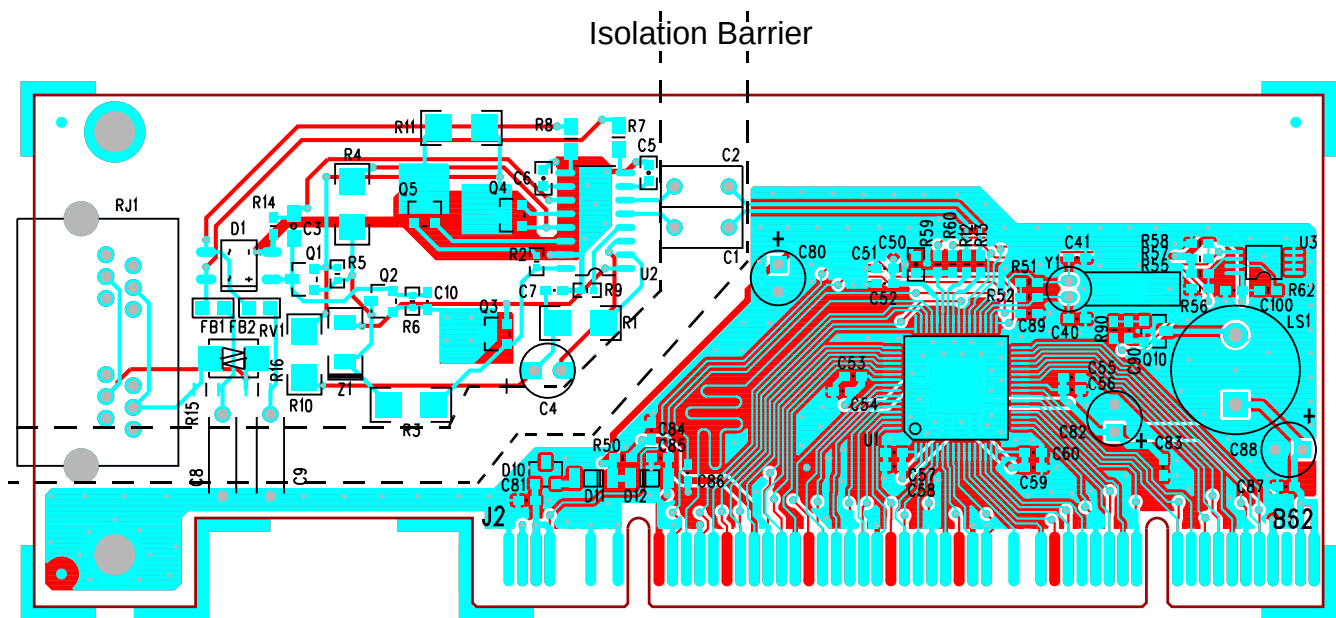


Figure 6. Si3052 Isolation Barrier

Modem-on-Motherboard or Custom PC Motherboard Riser: If possible, use through-hole Y2 class capacitors for the C1, C2, C8, and C9 components, and ensure a minimum of a 5 mm wide isolation barrier from TNV to SELV. A properly implemented design using these guidelines should provide a minimum of 5 kV of longitudinal surge immunity.

If surface mount components are required, use Y2 class, 22xx package, surface mount capacitors for isolation. Maintain an isolation barrier at least as wide as the spacing between the metal terminals of the capacitors. If possible, a 5 mm wide isolation barrier from TNV to SELV should be implemented in this case as well. A properly implemented design using these guidelines should provide a minimum of 5 kV of longitudinal surge immunity.

MDC/MiniPCI Modem: Use surface mount Y2 class capacitors for the C1, C2, C8, and C9 components, maintaining an isolation barrier at least as wide as the spacing between the metal terminals of the capacitors. To ensure a minimum of 3 kV longitudinal surge immunity, this spacing should be 3.5 mm or greater.

Note: Consult your safety expert for further information about the supplementary insulation requirements of your design.

Other Small Form-Factor Applications: Use surface mount Y2 class capacitors for the C1, C2, C8, and C9 components, maintaining an isolation barrier at least as wide as the spacing between the metal terminals of the capacitors. To ensure a minimum of 3 kV longitudinal surge immunity, this spacing should be 3.5 mm or greater. Customers may wish to build designs that can support different levels of surge immunity based on different customer requirements.

Note: Consult your safety expert for further information about the supplementary insulation requirements of your design in advance.

To ensure a minimum of 5 kV of surge immunity, use Y2 class, 22xx package, surface mount capacitors.

1.4. Si305x Power Supply Bypass Capacitor Layout Requirements

On the Si3050/54/56, one of the main layout considerations for the digital-side chip (U1) is the placement of the bypass capacitors (C50, C51, and, if required, C52). Capacitors C50 and C51 are the bypass capacitors for V_A and V_D , respectively. Figure 7 shows a typical placement of C50 and C51. The designer should focus on minimizing the length of the C50 connection between the V_A and GND pins and the length of the C51 connection between the V_D and GND pins on U1.

Figure 9 shows the bypass configuration of the Si3052 VIO pins. The Si3052 has several bypass capacitors that must be placed in close proximity to the VIO pins with short loops to GND. These capacitors use C53/C54, C55/C56, C57/C58, and C59/C60. In addition, the C50 capacitor on the V_A pin, and the C50/C51 capacitor on the V_D pin should be placed in close proximity to U1 with short loops to ground.

The Si3052 is connected to ground through the exposed paddle at the bottom of the package. To ensure the Si3052 package attaches reliably to the exposed paddle, a 5 mm square pad should be used. The solder paste should be laid out in a 4x4 matrix of 1 mm square pads on a 1.2 mm pitch (see Figure 8). It is recommended that a power pad be placed on the back of the Si3052 board directly behind the exposed die paddle pad. The combination of these two pads creates an additional bypass capacitor that assists with high-frequency bypassing of the Si3052 power supply.

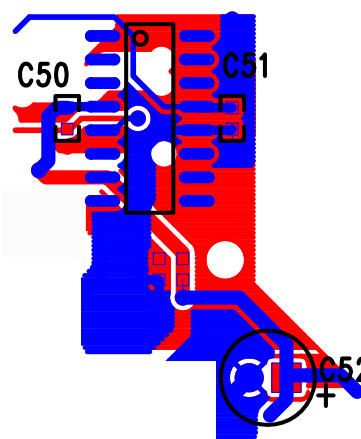


Figure 7. Si3050/54/56 Placement and Routing of C50–C52

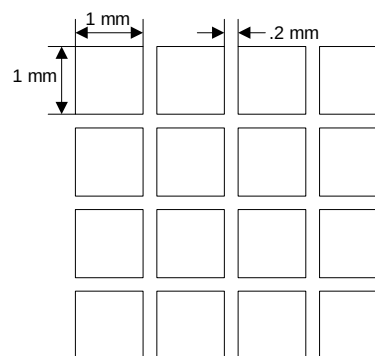


Figure 8. Si3052 Ground Pad Solder Paste Matrix on a 5x5 mm Pad

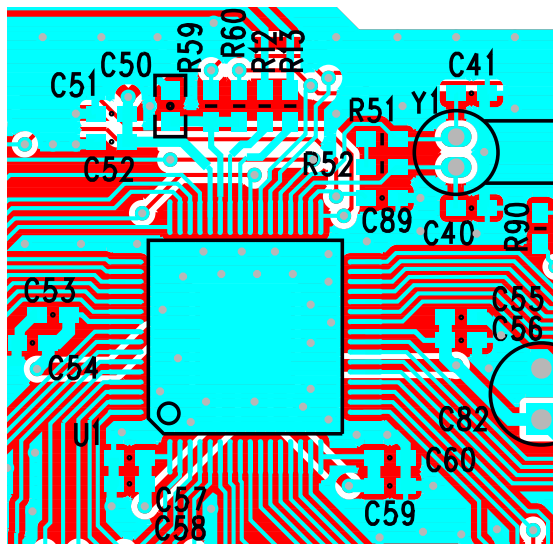


Figure 9. Si3052 Power Supply Bypassing

1.5. Si3052/54 Crystal Layout Requirements

The Si3052/54 incorporates a crystal interface for generation of a clock reference. For designs that use a crystal as shown in Figure 10, the following layout guidelines should be followed:

- The loop formed by XIN, Y1, and XOUT should be minimized and routed on one layer of the PCB.
- The loop formed by Y1, C40, and C41 should be minimized and routed on one layer of the PCB.
- The route from the GND pin to C40 and C41 should be as direct as possible and, if possible, on one layer of the PCB.

These guidelines also help minimize issues with emissions. An example layout is shown in Figures 11–13.

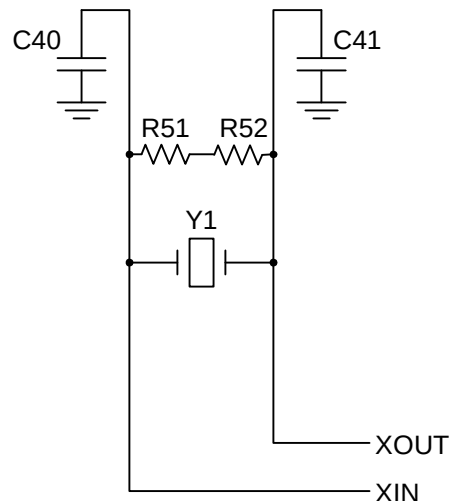


Figure 10. Crystal Circuit

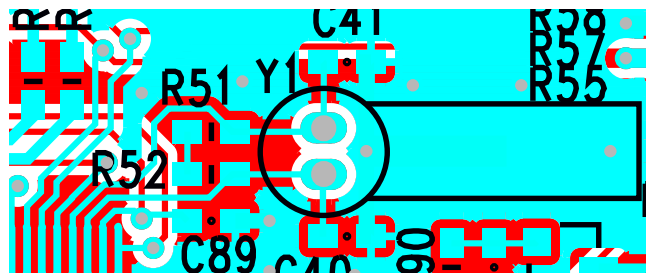


Figure 11. Si3052 PCI Y1 to U1 Routing

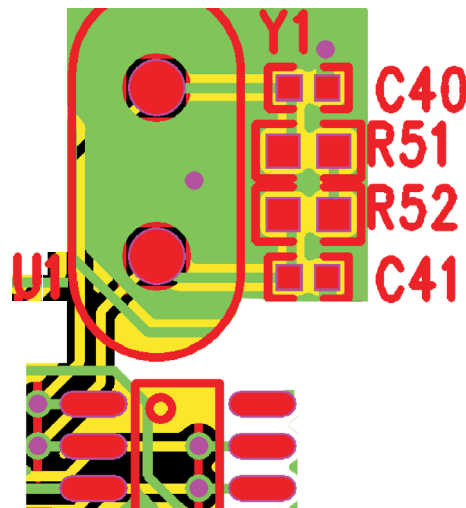


Figure 12. Si3054 CNR Y1 to U1 Routing

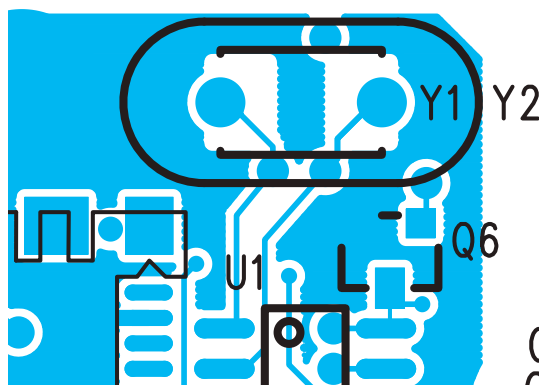


Figure 13. Si3054 MDC Y1 to U1 Routing

1.6. Si3018 Layout Recommendations

FB1, FB2, RV1, R15, R16, C8, and C9 should be placed as close as possible to the RJ11 as shown in Figures 14–16. Note that, for Si3050 designs, it is recommended that zero-ohm resistors R15 and R16 be replaced with ferrite beads in order to decrease emissions (see Si3050+Si3011/18/19 data sheet). The metallic surge protection device (RV1) should be located close to the RJ11 connector. It is important for the routing from the RJ11 connector through the ferrite beads FB1 and FB2 and the resistors R15 and R16 to be well matched. C8 and C9 should be placed so there is minimal distance between the nodes where they connect to chassis ground. The traces from the TIP and RING connections on the RJ11 through the EMC capacitors C8 and C9 to chassis ground should be kept as short as possible and should be well matched.

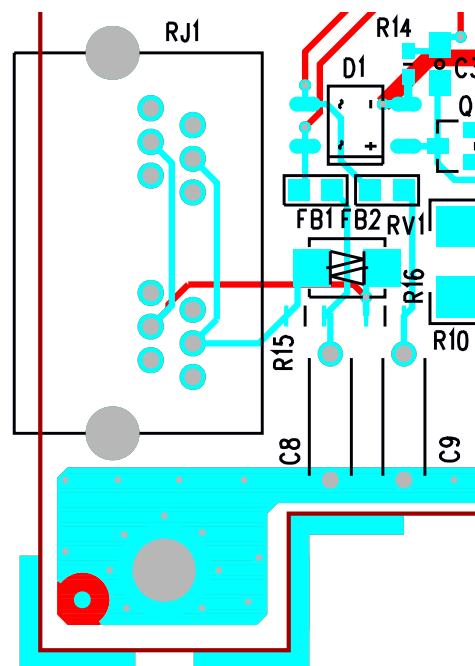


Figure 15. Si3052/54 PCI/CNR Placement and Routing of FB1, FB2, RV1, C8, C9, R15, and R16

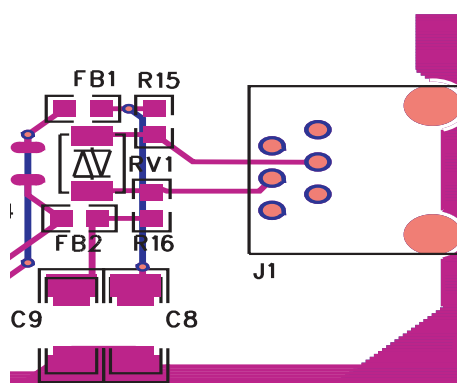


Figure 14. Si3050 Placement and Routing of FB1, FB2, RV1, C8, C9, R15, and R16



Figure 16. Si3054 MDC Placement and Routing of FB1, FB2, RV1, C8, C9, R15, and R16

After the Si305x system-side device, Si3018/19 line-side device, C1, and C2 have been placed, and the isolation barrier has been defined according to the recommendations described previously in this document, the designer should place the diode bridge and route the accompanying traces from the Si3018/19 device pin 11 through C3 to the Si3018/19 device and to the IGND pin. C3 should be placed across the diode bridge, and the area of the loop formed from Si3018/19 pin 11 through C3 to the diode bridge and back to Si3018/19 pin 15 should be minimized. An example of this routing for a variety of applications is shown in Figures 17–19.

Capacitors C5 and C6 provide regulation of the supplies powering U2. The loop formed by C5 to pins 15 and 7 and the loop formed by C6 to pins 15 and 10 should be made as small as possible. The trace back to pin 15 is thicker because multiple loops use this path.

Once these components have been placed, the designer can continue with the layout by adding Q4, Q5, R2, and R9. All traces connecting these components should be 15 mil in thickness. Place Q4 so that the area of the loop formed from U2 pin 13 to the base of Q4 and from U2 pin 12 to the emitter of Q4 is minimized. Place Q5 so that the loop from pin 16 through R2 to the base of Q5 and through the emitter to pin 14 is short. The loop formed from U2 pin 4 to R9 to U2 pin 15 should also be made as small as possible.

The IGND traces are those traces that connect directly from a capacitor or resistor to pin 15 of the line-side device. They should be routed on the PCB using 15 mil width traces.

Example layouts of these critical traces and components around the line-side device are shown in Figures 17–19:

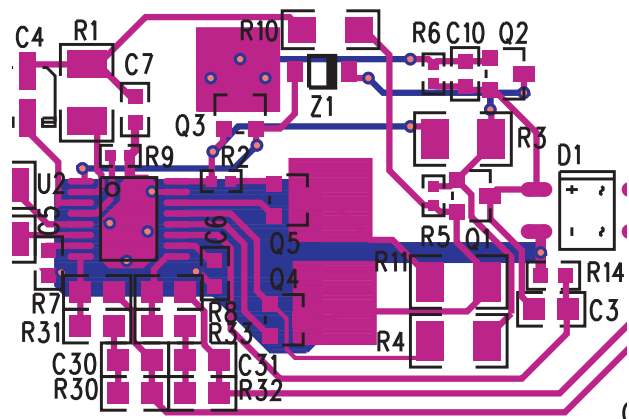


Figure 17. Si3050 Placement and Routing of C3, Diode Bridge, Q4, Q5, R9, R2, C5, and C6

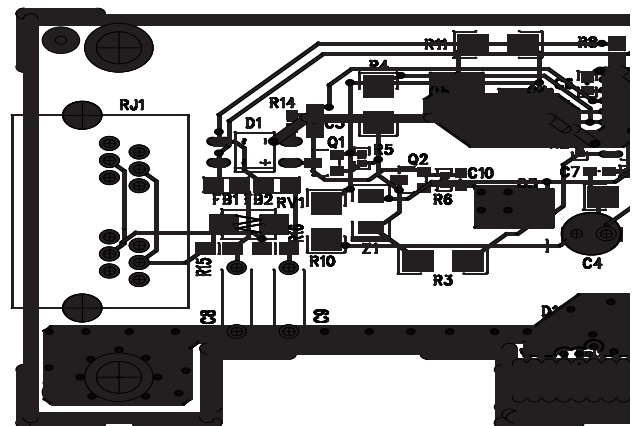


Figure 18. Si3052/54 PCI/CNR Placement and Routing of C3, Diode Bridge, Q4, Q5, R9, R2, C5, and C6

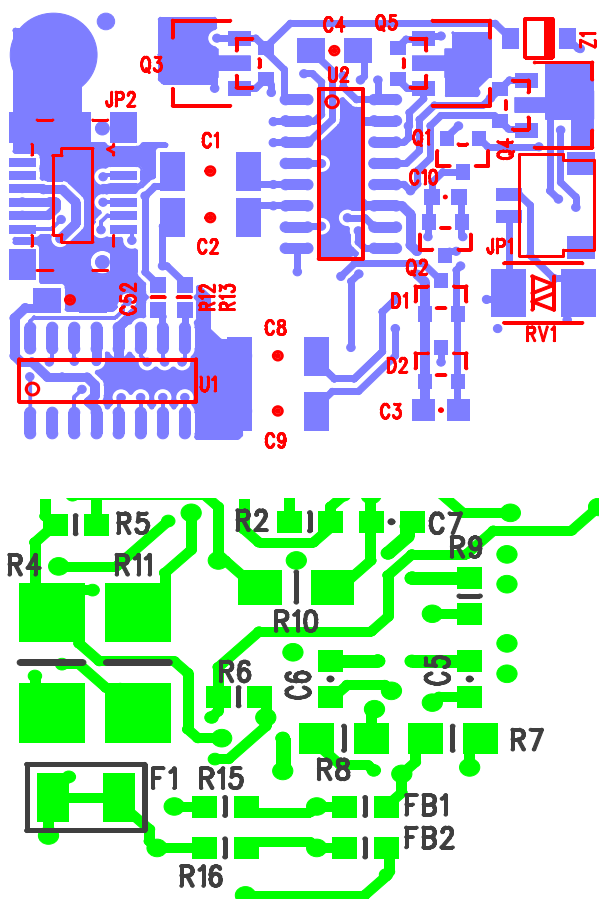


Figure 19. Si3054 MDC Placement and Routing of C3, Diode Bridge, Q4, Q5, R9, R2, C5, and C6

Finally, the layout can be completed by the addition of the remainder of the components in the line-side region and routing of the remainder of the signals. Although the layout considerations for these traces are not as stringent as the previous components and traces, it is recommended that the PCB designer follow these example placements and layouts as closely as possible. Example layouts in a variety of form factors are shown in Figures 20–22.

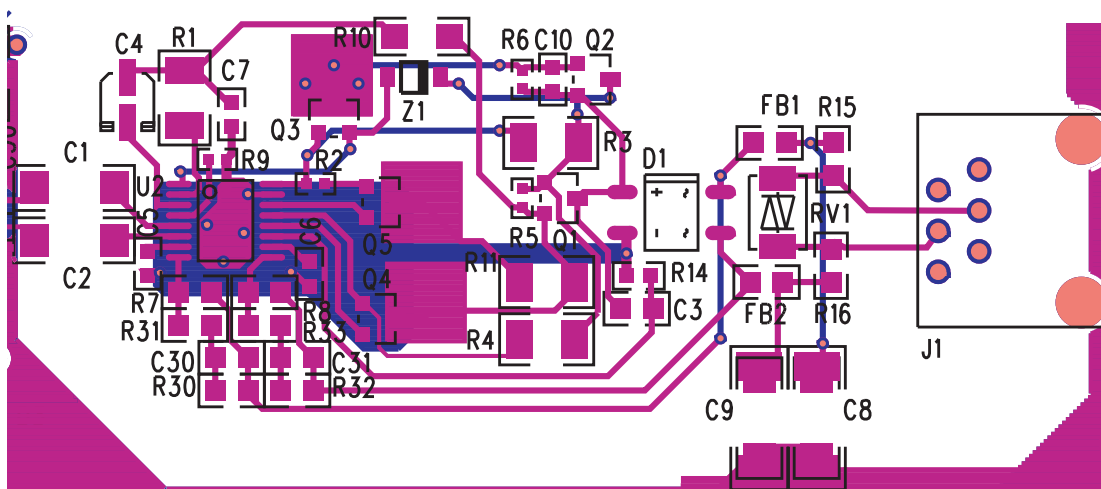


Figure 20. Si3050 Finished Layout Line-Side Portion

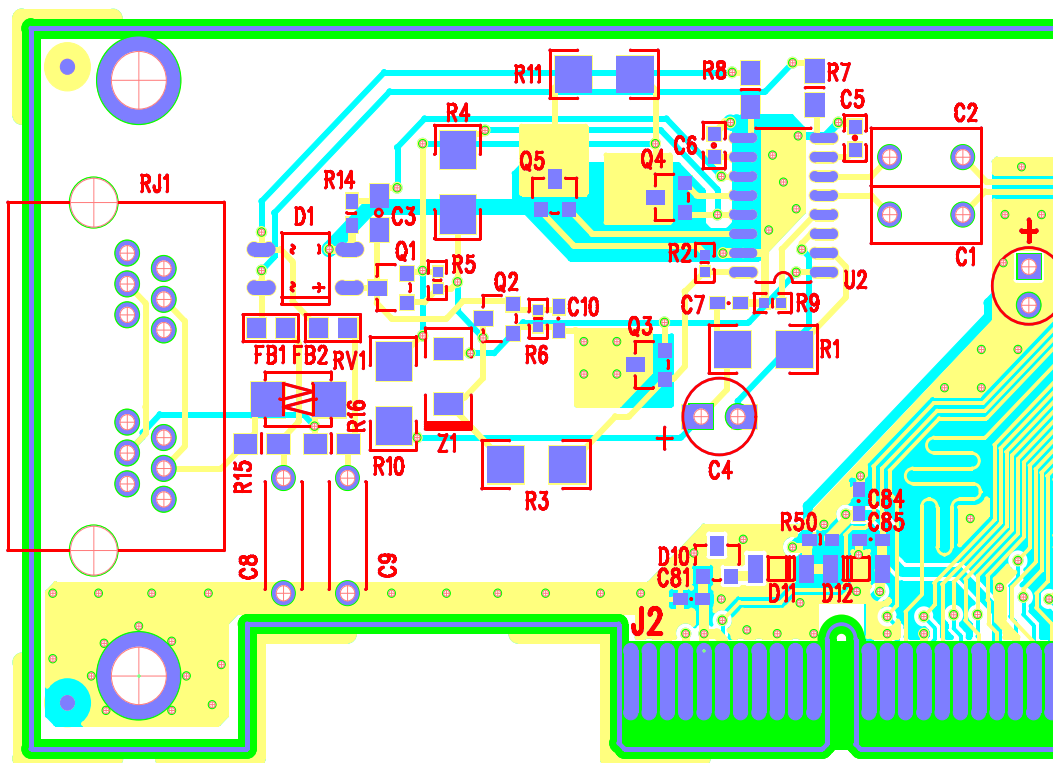
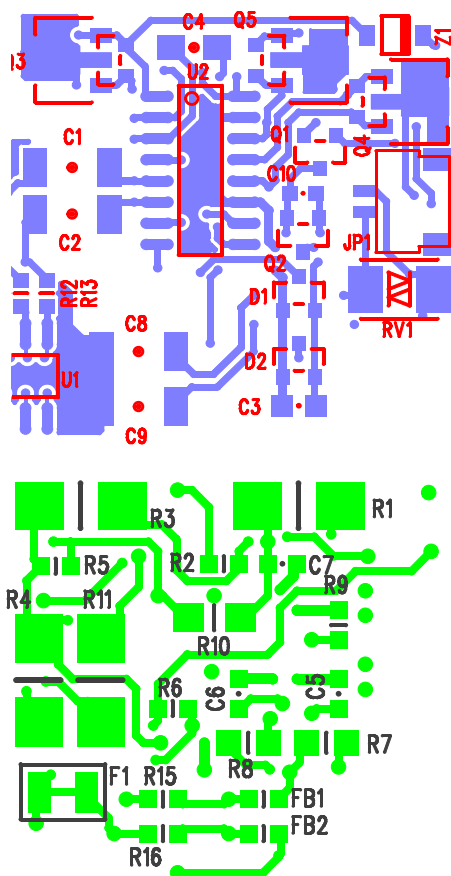


Figure 21. Si3052/54 PCI/CNR Finished Layout Line-Side Portion



**Figure 22. Si3054 MDC Finished Layout
Line-Side Portion**

1.7. Assembly Considerations

There are several steps that can be taken in layout to ensure that the assembly process is successful. An example of an assembly-related error is the installation of a polarized capacitor with the polarity backwards. Stenciling the board with a plus sign on the correct side of C4 if a polarized capacitor is used to indicate the proper orientation for that capacitor can prevent this.

Also, indicating pin 1 on the board with a stencil marking improves the chances that the integrated circuits will be installed correctly. The silkscreen for Z1 should also have a marking to indicate the cathode placement.

The designer should use several footprints for a given component to allow for multiple vendor choices. Popular components using multiple footprints are C1, C2, C8, C9, and Z1.

Finally, all 3-pin devices' orientation should be carefully checked to ensure that it matches the schematic symbol and routing implementation.

Taking these steps will assist in the assembly process and ease future troubleshooting and substitutions.

1.8. PCI Layout Considerations

The PCI Specification Revision 2.3 contains a section covering PCI board mechanical requirements. All PCI board designs should incorporate these recommendations. They can be found in the Revision 2.3 PCI Specification Section 4.4.3. The PCI Specification may be obtained from the PCI SIG at <http://www.pcisig.com/home>.

1.9. Small Form Factor Considerations

In small form factor applications, there are a number of considerations separate from other applications. An important consideration in small form factor designs is the heat dissipation under worst case conditions. Small form factor designs have less board mass to dissipate heat, so special precautions should be taken:

- The Skyworks Solutions DAA uses multiple components to distribute the power that every DAA must dissipate. This distribution of power among multiple components allows the layout designer the flexibility to space these components apart for a more thermally robust design.
- U2, Q3, Q4, Q5, R1, R3, R4, R10, and R11 should be spaced as far away from each other in the line-side section of the layout as possible. When designing a double-sided board, optimal thermal performance is achieved by ensuring that these high-power components are not placed directly underneath each other when placing them.
- Increase the size of the collector pads for Q3, Q4, and Q5 to improve the heat transfer from the die of these devices into the board.
- Fill the open area underneath the U2 package with IGND trace. An IGND pad should be placed on the back side directly underneath the body of the U2 package. Multiple vias should be used to stitch between the two pads to improve thermal transfer. This improves the heat transfer from the U2 die into the board.

Adding additional copper layers within the layers of the PCB can significantly improve the thermal performance of the design. A 4-layer board, with the inner layers underneath the line-side circuitry connected to IGND, is recommended.

The Skyworks Solutions' next generation DAA uses multiple components to distribute the power that every DAA, including other silicon DAAs and transformer-based architectures, must dissipate under certain line conditions.

The Skyworks Solutions architecture allows space to be designed between components and ensures the temperature of some components does not become too high. Temperature measurements on other DAA architectures that use only one or two components to dissipate the majority of the power have shown excessive temperature levels on some components. This can lead to performance and reliability issues. The Skyworks Solutions architecture does not exhibit this issue as long as careful component, layout, and placement considerations are made. Care should be taken, especially in MDC designs, due to the small form factor.

The higher power devices dissipate various amounts of power in different modes and under different loop conditions. In the worst case, the power dissipation for R1, R3, R4, R10, and R11 can be between 253 mW and approximately 450 mW, depending on which component is measured, what mode the DAA is in, and the line condition.

Skyworks Solutions' reference design recommends 2010 package resistors for R1, R3, R4, R10, and R11, which are usually rated as 1/2 W, depending on the supplier. In some cases, 1210 package devices have a 1/2 W rating, so they would be acceptable as well for 0–70 °C operation. Another option is to use two 1206 resistors (1/4 W rating each) in parallel with the correct parallel resistance combination to match the original value. These would have to be 1% tolerance, 1/4 W devices. R10 should be a minimum of 1/4 W rated device.

In some cases, 2010 package resistors are more widely available than 1210 package devices. That is the reason they are used in the Skyworks Solutions MDC reference design.

Q3, Q4, and Q5 also dissipate significant amounts of power in various modes under different line conditions, up to 180 mW (but not all at the same time). Those devices' maximum rated junction temperature is 150 °C. Calculations of the junction temperature can be made using the θ_{JA} values that the data sheets provide, but these are usually not very accurate due to variable testing conditions and different methods of measurement. Direct temperature measurements for these components provide a closer estimate of the junction temperature. Using this data along with the θ_{JC} or ψ_{JC} variables from the SOT-23 or SOT-89 transistor vendors, the junction temperature may be estimated. To obtain more thermal margin, use SOT-89 transistors for any of these three components.

Every board layout has different thermal performance. The design variables that affect thermal performance are as follows:

- Component package selection—Q3, Q4, Q5 SOT-23 or SOT 89, and R1, R3, R4, R10, R11.
- Component placement—Spacing these components far apart, with none of them directly underneath each other or the Si3018 if a double-sided design is implemented.
- Layout of traces—Using thick traces, with additional node-connected copper on the trace and inner layers improves the thermal performance of the Q3, Q4, Q5 components.
- Number of board layers—Using four layers has demonstrated improved performance over two layers, with IGND-connected fill on the inner layers of the line-side portion.

1.9.1. Small Form Factor TNV-SELV Isolation Barrier Creepage

Another difference in small form factor DAAs is the creepage distance from TNV to SELV. For applications that are targeted for global telecom regulatory compliance, Y2 class surface mount capacitors should be implemented for C1, C2, C8, and C9. These capacitors may be available in a variety of packages, including 1808, 2211, and 2220. Skyworks Solutions recommends that the minimum spacing from TNV to SELV for the isolation barrier on the rest of the board be derived from the spacing between the terminals of the isolation capacitors. For example, if 1808 Y2 class capacitors that have spacing between the terminals of 4.0 mm are used for C1, C2, C8, and C9, then the width of the isolation barrier between TNV and SELV circuitry everywhere else on the board should be a minimum of 4.0 mm.

Note: Consult your safety expert for further information about the supplementary insulation requirements of your design.

1.10. Si3050 Layout Check List

Table 1 is a checklist that the designer can use during the layout process to ensure all the recommendations in this application note have been implemented. Note that, for Si3050 designs, it is recommended that zero-ohm resistors R15 and R16 be replaced with ferrite beads in order to decrease emissions (see Si3050+Si3011/18/19 data sheet).

Table 1. Si3050 Layout Check List

✓	#	Layout Items	Required
	1	U1 and U2 are placed so that pins 11–20 of U1 are facing pins 1–8 of U2. C1 and C2 are placed directly between U1 and U2. Keep R12 and R13 close to U1.	
	2	Place U1, U2, C1, and C2 so that the recommended minimum creepage spacing for the target application is implemented. See the section entitled 1.3."Isolation Barrier Creepage Spacing Considerations" on page 4.	
	3	C1 and C2 should be placed directly between U1 and U2. Short, direct traces should be used to connect C1 and C2 to U1 and U2. These traces should never be longer than two inches and should be minimized in length. Place C2 such that its accompanying trace to the C2B pin (pin 6) on the Si3019 is not close to the trace from R31 to the RNG1 pin on the Si3019 (pin 8).	
	4	Place R30–R33, and C30–C31 as close as possible to the RNG1 and RNG2 pins (pins 8 and 9), ensuring a minimum trace length from the RNG1 or RNG2 pin to the R31 or R33 resistor. In order to space the R31 component further from the trace from C2 to the C2B pin, it is acceptable to orient it 90 degrees relative to the RNG1 pin (pin 8).	
	5	The area of the loop from C50 to U1 pin 17 and from C51 to pin 16 back to pin 18 (DGND) should be minimized. The return traces to U1 pin 18 (DGND) should be on the component side.	
	6	The digital ground plane is made as small as possible, and the ground plane has rounded corners.	
	7	Use a minimum of 15 mil width traces in DAA section, use a minimum of 20 mil width traces for IGND.	
	8	C3 should be placed across the diode bridge, and the area of the loop formed from Si3019 pin 11 through C3 to the diode bridge and back to Si3019 pin 15 should be minimized.	
	9	FB1, FB2, R15, R16, and RV1 should be placed as close as possible to the RJ11.	
	10	C8 and C9 should be placed so that there is a minimal distance between the nodes where they connect to chassis ground.	
	11	Use a minimum of 20 mil wide trace from RJ11 to FB1, FB2, R15, R16, RV1, C8, C9, and F1.	
	12	The routing from TIP and RING of the RJ11 through the ferrite beads should be well matched.	
	13	The traces from the RJ11 through the Ringer Network to U2 pin 8 and pin 9 should be well matched. These traces may be up to 10 cm long.	
	14	Distance from TIP and RING through EMC capacitors C8 and C9 to chassis ground is short.	

Table 1. Si3050 Layout Check List (Continued)

✓	#	Layout Items	Required
	15	There should be no digital ground plane in the DAA Section.	
	16	Minimize the area of the loop from U2 pin 7 and pin 10 to C5 and C6 and from those components to U2 pin 15 (IGND).	
	17	R2 should be placed next to the base of Q5, and the trace from R2 to U2 pin16 should be less than 20 mm.	
	18	Place C4 close to U2 and connect C4 to U2 using a short, direct trace.	
	19	The area of the loop formed from U2 pin 13 to the base of Q4 and from U2 pin 12 to the emitter of Q4 should be minimized.	
	20	The trace from C7 to U2 pin 15 should be short and direct.	
	21	The trace from C3 to the D1/D2 node should be short and direct.	
	22	Provide a minimum of 5 mm creepage (or use the capacitor terminal plating spacing as a guideline for small form factor applications) from any TNV component, pad or trace to any SELV component, pad or trace.	
	23	Minimize the area of the loop formed from U2 pin 4 to R9 to U2 pin 15.	
	24	Cathode marking for Z1.	
	25	Pin 1 marking for U1 and U2.	
	26	Space and mounting holes to accommodate for fire enclosure if necessary.	
Small Form Factor Specific Guidelines			
	27	Space U2, Q4, Q5, R1, R3, R4, R10, and R11 far away from each other in the line-side section of the layout.	
	28	Make the size of the collector pads for Q3–Q5 larger to improve heat dissipation. Implement collector pads on the back side of the board and connect the two with multiple vias to improve heat transfer.	
	29	TNV to SELV creepage matches the spacing between the surface mount capacitor terminals.	
	30	The area underneath the body of U2 should be filled with IGND trace, and an IGND pad should be placed on the back side directly underneath the body of the U2 package. Multiple vias should be used to stitch between the two pads to improve thermal transfer.	
	31	Using IGND planes on inner layers is useful in attaining uniform heat transfer and dissipation.	



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1.11. Si3052 PCI Layout Check List

Table 2 is a checklist that the designer can use during the layout process to ensure all the recommendations in this application note have been implemented.

Table 2. Si3052 PCI Layout Check List

✓	#	Layout Items	Required
	1	Place U1, U2, C1, and C2 so that the recommended minimum creepage spacing for the target application is implemented. Keep R12 and R13 close to U1. See the section entitled 1.3."Isolation Barrier Creepage Spacing Considerations" on page 4.	
	2	C1 and C2 should be placed directly between U1 and U2. Short, direct traces should be used to connect C1 and C2 to U1 and U2. These traces should never be longer than two inches and should be minimized in length. Place C2 such that its accompanying trace to the C2B pin (pin 6) on the Si3018 is not close to the trace from R7 to the RNG1 pin on the Si3018 (pin 8). R12 and R13 should be placed close to U1. R12 and R13 should be placed close to U1.	
	3	Place R7 and R8 as close as possible to the RNG1 and RNG2 pins (pins 8 and 9), ensuring a minimum trace length from the RNG1 or RNG2 pin to the R7 or R8 resistor. In order to space the R7 component further from the trace from C2 to the C2B pin, it is acceptable to orient it 90 degrees relative to the RNG1 pin (pin 8).	
	4	The area of the loop from C50 to U1 pin 43 and from C51/C52 to pin 44 back to DGND should be minimized. The return traces to U1 DGND should be on the component side.	
	5	The loop formed by XIN, Y1, and XOUT should be minimized and routed on one layer. The loop formed by Y1, C40, and C41 should be minimized and routed on one layer.	
	7	Use a minimum of 15 mil width traces in DAA section, use a minimum of 20 mil width traces for IGND.	
	8	C3 should be placed across the diode bridge, and the area of the loop formed from Si3018/19 pin 11 through C3 to the diode bridge and back to Si3018/19 pin 15 should be minimized.	
	9	FB1, FB2, R15, R16, and RV1 should be placed as close as possible to the RJ11.	
	10	C8 and C9 should be placed so that there is a minimal distance between the nodes where they connect to chassis ground.	
	11	Use a minimum of 20 mil wide trace from RJ11 to FB1, FB2, RV1, C8, C9.	
	12	The routing from TIP and RING of the RJ11 to the ferrite beads should be well matched.	
	13	The traces from the RJ11 through R7 and R8 to U2 pin 8 and pin 9 should be well matched. These traces may be up to 10 cm long.	
	14	Distance from TIP and RING through EMC capacitors C8 and C9 to chassis ground is short.	
	15	There should be no digital ground plane in the DAA Section.	
	16	Minimize the area of the loop from U2 pin 7 and pin 10 to C5 and C6 and from those components to U2 pin 15 (IGND).	

Table 2. Si3052 PCI Layout Check List (Continued)

✓	#	Layout Items	Required
	17	R2 should be placed next to the base of Q5, and the trace from R2 to U2 pin16 should be less than 20 mm.	
	18	Place C4 close to U2 and connect C4 to U2 using a short, direct trace.	
	19	The area of the loop formed from U2 pin 13 to the base of Q4 and from U2 pin 12 to the emitter of Q4 should be minimized.	
	20	The trace from C7 to U2 pin 15 should be short and direct.	
	21	The trace from C3 to the D1/D2 node should be short and direct.	
	22	Provide a minimum of 5 mm creepage (or use the capacitor terminal plating spacing as a guideline for small form factor applications) from any TNV component, pad or trace to any SELV component, pad or trace.	
	23	Minimize the area of the loop formed from U2 pin 4 to R9 to U2 pin 15.	
	24	Cathode marking for Z1.	
	25	Pin 1 marking for U1 and U2.	
	26	Space and mounting holes to accommodate for fire enclosure if necessary.	
	27	Create a power pad on the back side of the board beneath U1 to increase noise immunity on the power supply.	
	28	Place Si3052 V_A , V_D , and V_{IO} coupling caps close to U1.	
	29	Make the size of the collector pads for Q3, Q4, and Q5 larger to improve heat dissipation.	
	31	Solder mask for the exposed paddle on U1 should be as shown in Figure 8 on page 5.	
	32	Place a power supply pad directly underneath the exposed paddle to create a high-frequency bypass capacitor for U1.	

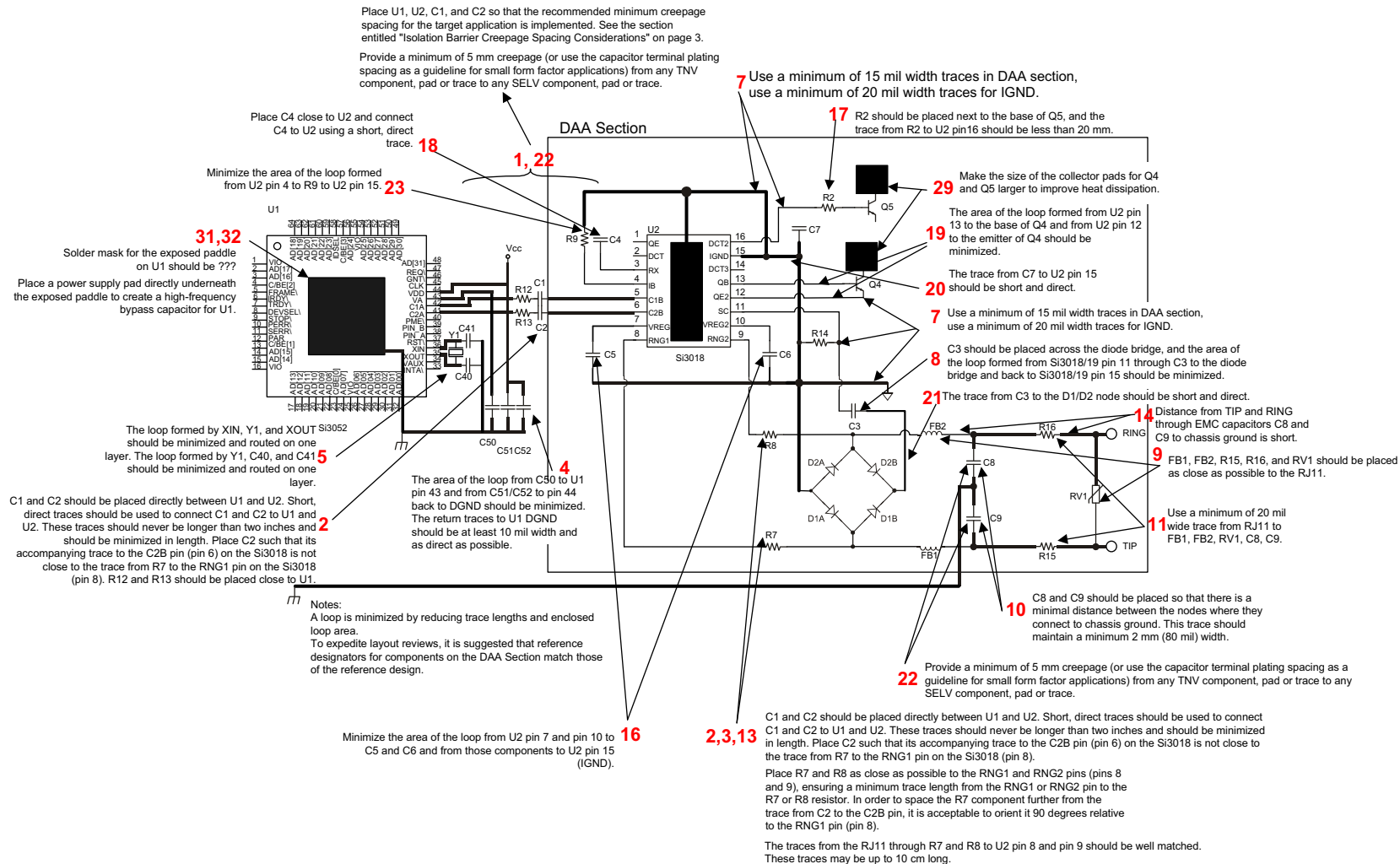


Figure 24. Si3052 PCI Layout Guidelines

1.12. Si3054 CNR/ACR/AMR Layout Check List

Table 3 is a checklist that the designer can use during the layout process to ensure all the recommendations in this application note have been implemented. Additionally, Figure 25 provides an annotated diagram of all of the relevant layout guidelines for the Si3054 CNR/AMR/ACR applications.

Table 3. Si3054 CNR/ACR/AMR Layout Check List

✓	#	Layout Items	Required
	1	U1 and U2 are placed so that pins 9–16 of U1 are facing pins 1–8 of U2. C1 and C2 are placed directly between U1 and U2.	
	2	Place U1, U2, C1, and C2 so that the recommended minimum creepage spacing for the target application is implemented. See the section entitled 1.3."Isolation Barrier Creepage Spacing Considerations" on page 4. R12 and R13 should be close to U1.	
	3	C1 and C2 should be placed directly between U1 and U2. Short, direct traces should be used to connect C1 and C2 to U1 and U2. These traces should never be longer than two inches and should be minimized in length. Place C2 such that its accompanying trace to the C2B pin (pin 6) on the Si3018 is not close to the trace from R7 to the RNG1 pin on the Si3018 (pin 8).	
	4	Place R7 and R8 as close as possible to the RNG1 and RNG2 pins (pins 8 and 9), ensuring a minimum trace length from the RNG1 or RNG2 pin to the R7 or R8 resistor. In order to space the R7 component further from the trace from C2 to the C2B pin, it is acceptable to orient it 90 degrees relative to the RNG1 pin (pin 8).	
	5	The area of the loop from C50 to U1 pin 4 and from C51 to pin 13 back to pin 12 (DGND) should be minimized. The return traces to U2 pin 12 (DGND) should be on the component side.	
	6	The loop formed by XIN, Y1, and XOUT should be minimized and routed on one layer. The loop formed by Y1, C40, and C41 should be minimized and routed on one layer.	
	7	The digital ground plane is made as small as possible, and the ground plane has rounded corners.	
	8	Series resistors on clock signals are placed near source.	
	9	Use a minimum of 15 mil width traces in DAA section, use a minimum of 20 mil width traces for IGND.	
	10	C3 should be placed across the diode bridge, and the area of the loop formed from Si3018/19 pin 11 through C3 to the diode bridge and back to Si3018/19 pin 15 should be minimized.	
	11	FB1, FB2, and RV1 should be placed as close as possible to the RJ11.	
	12	C8 and C9 should be placed so that there is a minimal distance between the nodes where they connect to chassis ground.	
	13	Use a minimum of 20 mil wide trace from RJ11 to FB1, FB2, RV1, C8, C9, and F1.	
	14	The routing from TIP and RING of the RJ11 through F1 to the ferrite beads should be well matched.	

Table 3. Si3054 CNR/ACR/AMR Layout Check List (Continued)

✓	#	Layout Items	Required
	15	The traces from the RJ11 through R7 and R8 to U2 pin 8 and pin 9 should be well matched. These traces may be up to 10 cm long.	
	16	Distance from TIP and RING through EMC capacitors C8 and C9 to chassis ground is short.	
	17	There should be no digital ground plane in the DAA Section.	
	18	Minimize the area of the loop from U2 pin 7 and pin 10 to C5 and C6 and from those components to U2 pin 15 (IGND).	
	19	R2 should be placed next to the base of Q5, and the trace from R2 to U2 pin16 should be less than 20 mm.	
	20	Place C4 close to U2 and connect C4 to U2 using a short, direct trace.	
	21	The area of the loop formed from U2 pin 13 to the base of Q4 and from U2 pin 12 to the emitter of Q4 should be minimized.	
	22	The trace from C7 to U2 pin 15 should be short and direct.	
	23	The trace from C3 to the D1/D2 node should be short and direct.	
	24	Provide a minimum of 5 mm creepage (or use the capacitor terminal plating spacing as a guideline for small form factor applications) from any TNV component, pad or trace to any SELV component, pad or trace.	
	25	Minimize the area of the loop formed from U2 pin 4 to R9 to U2 pin 15.	
	26	Cathode marking for Z1.	
	27	Pin 1 marking for U1 and U2.	
	28	Space and mounting holes to accommodate for fire enclosure if necessary.	

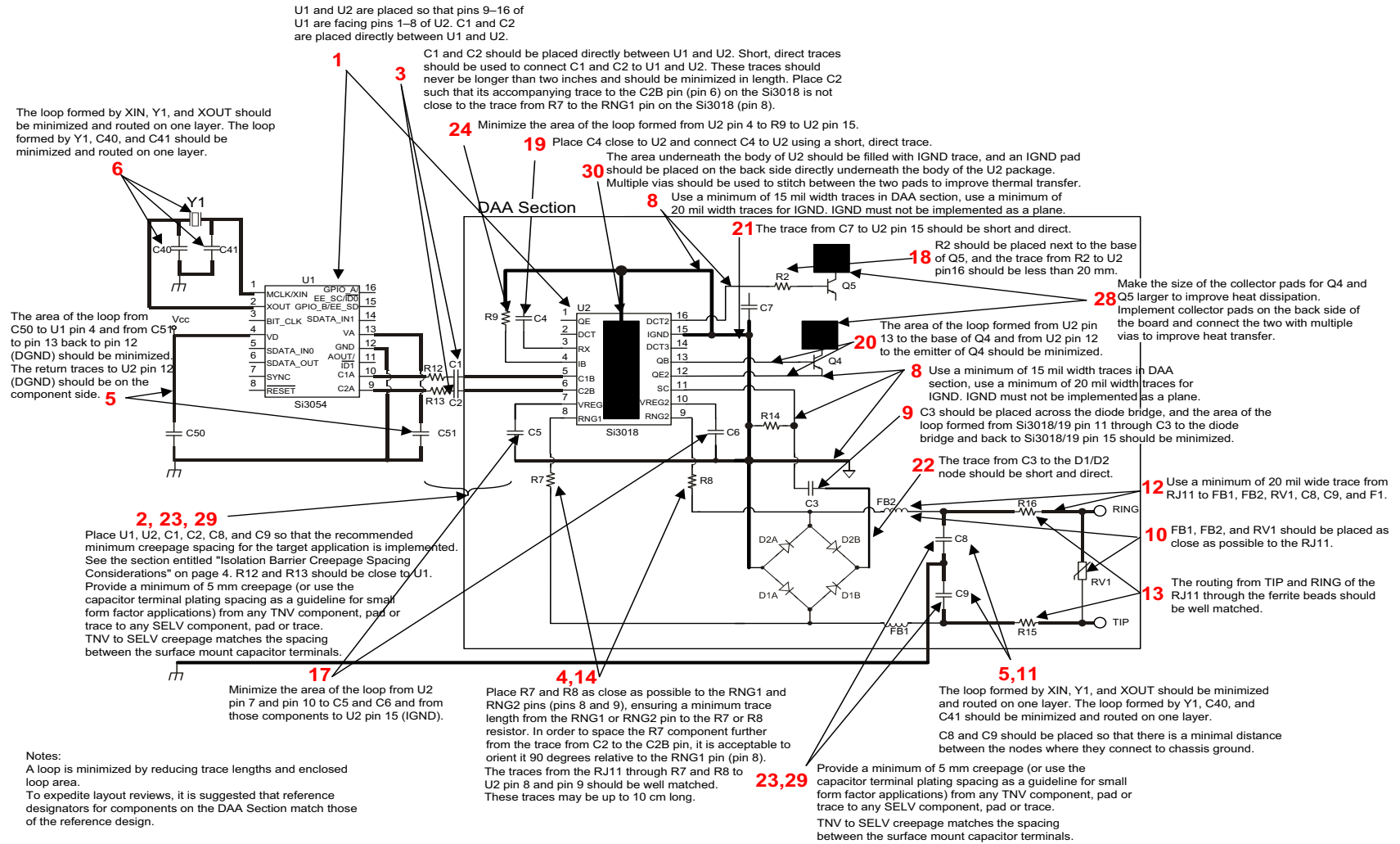


Figure 25. Si3054 CNR/ACR/AMR Layout Guidelines

1.13. Si3054 MDC/miniPCI Layout Check List

Table 4 is a checklist that the designer can use during the layout process to ensure all the recommendations in this application note have been implemented.

Table 4. Si3054 MDC/miniPCI Layout Check List

✓	#	Layout Items	Required
	1	U1 and U2 are placed so that pins 9–16 of U1 are facing pins 1–8 of U2. C1 and C2 are placed directly between U1 and U2. R12 and R13 should be placed close to U1.	
	2	Place U1, U2, C1, C2, C8, and C9 so that the recommended minimum creepage spacing for the target application is implemented. See the section entitled 1.3."Isolation Barrier Creepage Spacing Considerations" on page 4.	
	3	C1 and C2 should be placed directly between U1 and U2. Short, direct traces should be used to connect C1 and C2 to U1 and U2. These traces should never be longer than two inches and should be minimized in length. Place C2 such that its accompanying trace to the C2B pin (pin 6) on the Si3018 is not close to the trace from R7 to the RNG1 pin on the Si3018 (pin 8).	
	4	Place R7 and R8 as close as possible to the RNG1 and RNG2 pins (pins 8 and 9), ensuring a minimum trace length from the RNG1 or RNG2 pin to the R7 or R8 resistor. In order to space the R7 component further from the trace from C2 to the C2B pin, it is acceptable to orient it 90 degrees relative to the RNG1 pin (pin 8).	
	5	The area of the loop from C50 to U1 pin 4 and from C51 to pin 13 back to pin 12 (DGND) should be minimized. The return traces to U2 pin 12 (DGND) should be on the component side.	
	6	The loop formed by XIN, Y1, and XOUT should be minimized and routed on one layer. The loop formed by Y1, C40, and C41 should be minimized and routed on one layer.	
	7	Series resistors on clock signals are placed near source.	
	8	Use a minimum of 15 mil width traces in DAA section, use a minimum of 20 mil width traces for IGND. IGND must not be implemented as a plane.	
	9	C3 should be placed across the diode bridge, and the area of the loop formed from Si3018/19 pin 11 through C3 to the diode bridge and back to Si3018/19 pin 15 should be minimized.	
	10	FB1, FB2, and RV1 should be placed as close as possible to the RJ11.	
	11	C8 and C9 should be placed so that there is a minimal distance between the nodes where they connect to chassis ground.	
	12	Use a minimum of 20 mil wide trace from RJ11 to FB1, FB2, RV1, C8, C9, and F1.	
	13	The routing from TIP and RING of the RJ11 through the ferrite beads should be well matched.	
	14	The traces from the RJ11 through R7 and R8 to U2 pin 8 and pin 9 should be well matched. These traces may be up to 10 cm long.	
	15	Distance from TIP and RING through EMC capacitors C8 and C9 to chassis ground is short.	

Table 4. Si3054 MDC/miniPCI Layout Check List (Continued)

✓	#	Layout Items	Required
	16	There should be no digital ground plane in the DAA Section.	
	17	Minimize the area of the loop from U2 pin 7 and pin 10 to C5 and C6 and from those components to U2 pin 15 (IGND).	
	18	R2 should be placed next to the base of Q5, and the trace from R2 to U2 pin16 should be less than 20 mm.	
	19	Place C4 close to U2 and connect C4 to U2 using a short, direct trace.	
	20	The area of the loop formed from U2 pin 13 to the base of Q4 and from U2 pin 12 to the emitter of Q4 should be minimized.	
	21	The trace from C7 to U2 pin 15 should be short and direct.	
	22	The trace from C3 to the D1/D2 node should be short and direct.	
	23	Provide a minimum of 5 mm creepage (or use the capacitor terminal plating spacing as a guideline for small form factor applications) from any TNV component, pad or trace to any SELV component, pad or trace.	
	24	Minimize the area of the loop formed from U2 pin 4 to R9 to U2 pin 15.	
	25	Cathode marking for Z1.	
	26	Pin 1 marking for U1 and U2.	
Small Form Factor Specific Guidelines			
	27	Space U2, Q4, Q5, R1, R3, R4, R10, and R11 far away from each other in the line-side section of the layout.	
	28	Make the size of the collector pads for Q4 and Q5 larger to improve heat dissipation. Implement collector pads on the back side of the board and connect the two with multiple vias to improve heat transfer.	
	29	TNV to SELV creepage matches the spacing between the surface mount capacitor terminals.	
	30	The area underneath the body of U2 should be filled with IGND trace, and an IGND pad should be placed on the back side directly underneath the body of the U2 package. Multiple vias should be used to stitch between the two pads to improve thermal transfer.	

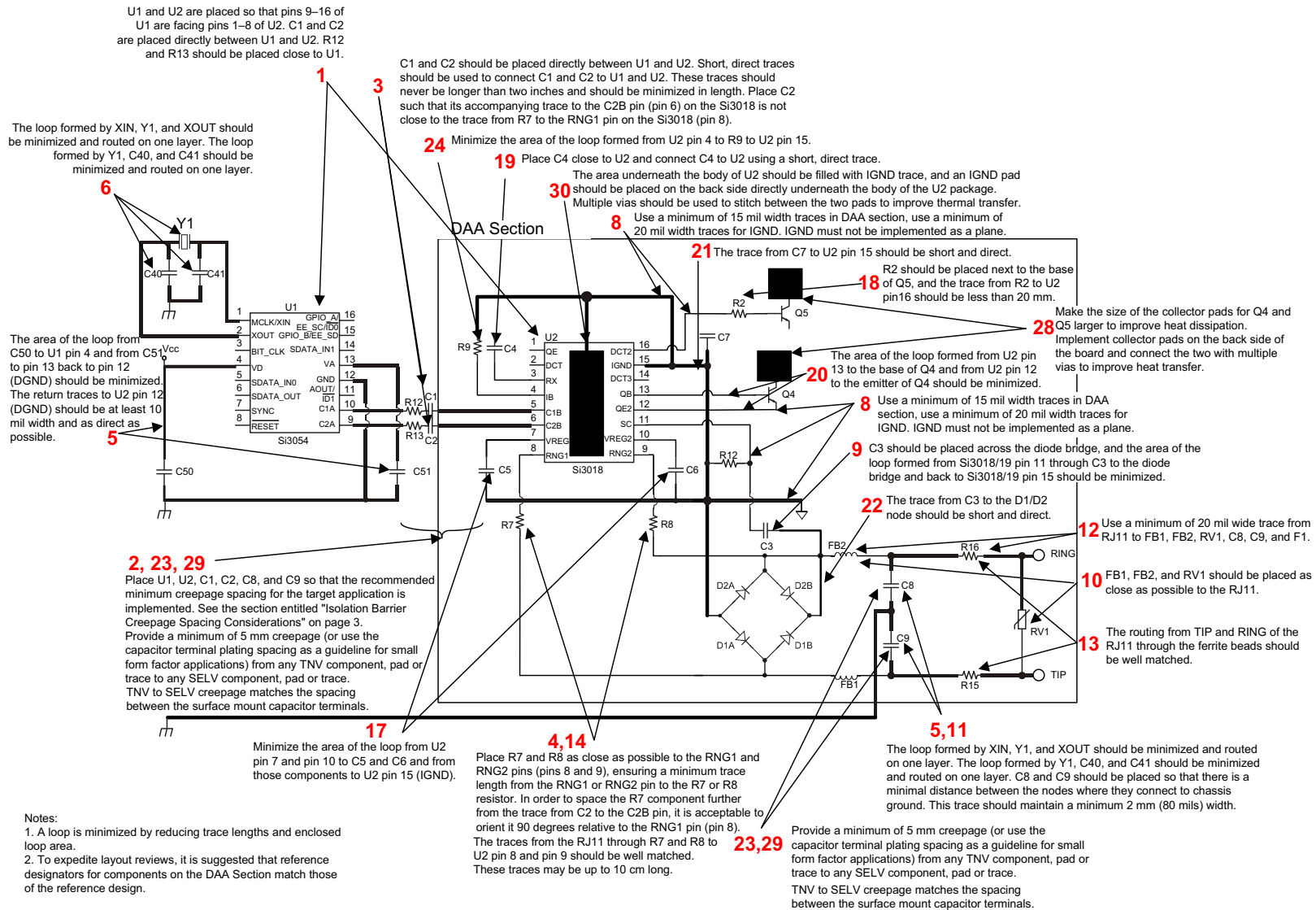


Figure 26. Si3054 MDC/miniPCI Layout Guidelines

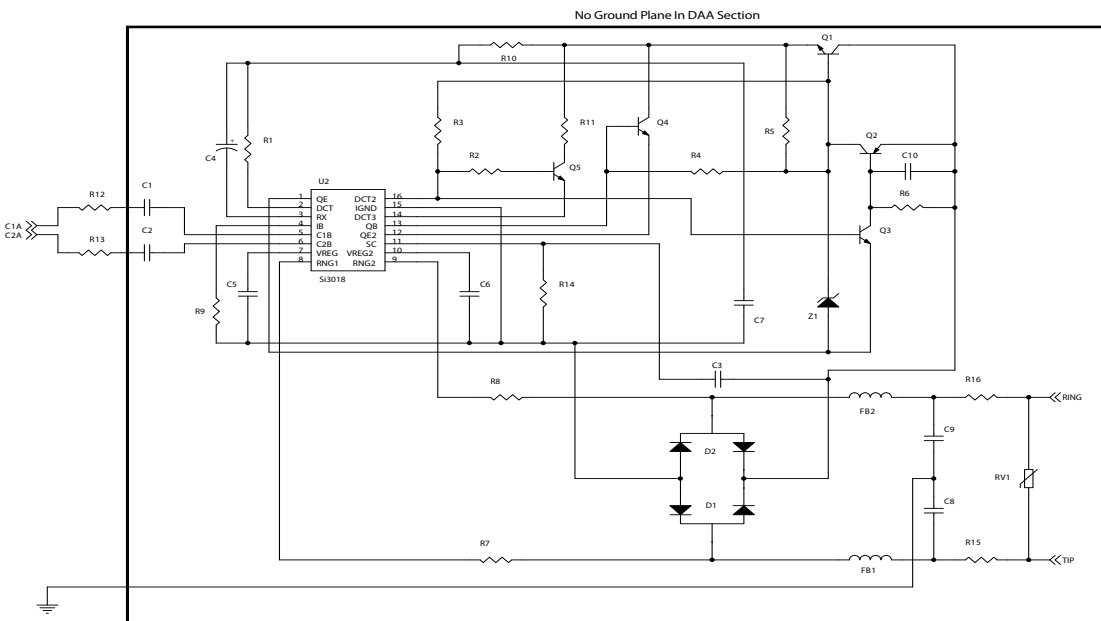


Figure 27. Si3050 Typical Application Schematic

DOCUMENT CHANGE LIST

Revision 0.2 to Revision 0.3

- Corrected C7 capacitor in Figure 23 on page 15.

Revision 0.3 to Revision 0.4

- Updated 1.1."Form Factor Overview" on page 1.
- Updated 1.2."Placement Guidelines" on page 1.
- Updated 1.6."Si3018 Layout Recommendations" on page 7.
- Updated 1.10."Si3050 Layout Check List" on page 13,

Revision 0.4 to Revision 0.5

- Removed references to pending 2005 update of EN60950.

NOTES:

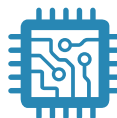


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